DAY 01: SMALL OFFICE NETWORK DESIGN AND IMPLEMENTATION

Project Category: Network Engineering
Project Type: LAN Design, VLAN Implementation and Inter-VLAN Routing
Simulation Tool: Cisco Packet Tracer
Router: Cisco 2911
Switch: Cisco 2960
Status: Completed

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1. Project Overview

This project involved designing and implementing a simulated Local Area Network (LAN) for a small office environment using Cisco Packet Tracer.

The office was divided into two departments: Human Resources (HR) and Information Technology (IT). Each department was assigned a separate logical network using Virtual Local Area Networks (VLANs).

The primary objective was to establish logical network segmentation while maintaining communication between the departments through inter-VLAN routing.

The network was implemented using a Cisco 2911 router, a Cisco 2960 switch and four end-user PCs.

2. Project Objectives

The project was designed to achieve the following objectives:

- Design a functional small-office LAN.
- Create and configure VLANs for two departments.
- Assign switch access ports to the appropriate VLANs.
- Configure trunking between the switch and router.
- Implement Router-on-a-Stick for inter-VLAN communication.
- Configure IPv4 addresses and default gateways.
- Verify network connectivity through ICMP ping tests.
- Understand the practical application of network segmentation and routing.

3. Network Requirements

The simulated office required:

Requirement	Description
Departments	HR and IT
Network segments	Two VLANs
End devices	Four PCs
Router	Cisco 2911
Switch	Cisco 2960
Routing	Inter-VLAN routing
Addressing	Static IPv4
Connectivity	Communication within and between VLANs

4. Network Topology and Architecture

The network follows a hierarchical star arrangement, with the router connected to the central switch. All four PCs connect to the switch through dedicated access ports.

The router provides communication between the two VLANs using separate logical subinterfaces on its physical GigabitEthernet interface.

Device connections

Device	Interface	Connected device	Interface
R1	G0/0	SW1	G0/1
SW1	Fa0/1	HR-PC1	FastEthernet 0
SW1	Fa0/2	HR-PC2	FastEthernet 0
SW1	Fa0/3	IT-PC1	FastEthernet 0

SW1	Fa0/4	IT-PC2	FastEthernet 0

Logical architecture

- VLAN 10 represents the Human Resources department.
- VLAN 20 represents the Information Technology department.
- The switch separates departmental traffic into the two VLANs.
- The router handles communication between the separate IP networks.
- The connection between SW1 and R1 operates as an IEEE 802.1Q trunk.

5. IP Addressing Scheme

The network uses private IPv4 addressing with a /24 subnet mask.

Device	Department	VLAN	IP Address	Subnet Mask	Default Gateway
HR-PC1	HR	10	192.168.10.10	255.255.255.0	192.168.10.1
HR-PC2	HR	10	192.168.10.11	255.255.255.0	192.168.10.1
IT-PC1	IT	20	192.168.20.10	255.255.255.0	192.168.20.1
IT-PC2	IT	20	192.168.20.11	255.255.255.0	192.168.20.1

Router interface addressing

Router interface	VLAN	IP address	Subnet mask
G0/0.10	10	192.168.10.1	255.255.255.0
G0/0.20	20	192.168.20.1	255.255.255.0

The router subinterface addresses serve as the default gateways for their respective VLANs.

6. VLAN Design and Port Allocation

Two VLANs were created to logically separate the office departments.

VLAN ID	VLAN Name	Assigned Switch Ports	Purpose
10	HR	Fa0/1–Fa0/2	Human Resources
20	IT	Fa0/3–Fa0/4	Information Technology

GigabitEthernet0/1 was configured as a trunk port to carry traffic for VLANs 10 and 20 between the switch and router.

7. Implementation and Configuration

7.1 Switch Configuration

The switch was configured with the hostname SW1. VLANs 10 and 20 were created, and the appropriate access ports were assigned to each VLAN.

The following commands were used:

```
None
enable
configure terminal
hostname SW1

vlan 10
name HR
exit

vlan 20
name IT
exit

interface range fastEthernet 0/1-2
switchport mode access
switchport access vlan 10
exit

interface range fastEthernet 0/3-4
```

```
switchport mode access
switchport access vlan 20
exit

interface gigabitEthernet 0/1
switchport mode trunk
switchport trunk allowed vlan 10,20
no shutdown
exit

end
write memory
```

Configuration explanation:

- `hostname SW1` identifies the switch.
- `vlan 10` and `vlan 20` create the departmental VLANs.
- `switchport mode access` configures end-device ports as access ports.
- `switchport access vlan` assigns each port to its designated VLAN.
- `switchport mode trunk` enables trunk operation.
- `switchport trunk allowed vlan 10,20` permits the two VLANs across the trunk.
- `write memory` saves the configuration.

7.2 Router Configuration

The router was configured using the Router-on-a-Stick approach.

A single physical interface, GigabitEthernet0/0, was divided into two logical subinterfaces.

```
None
enable
configure terminal
hostname R1

interface gigabitEthernet 0/0
no shutdown
exit
```

```
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet 0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit

end
write memory
```

Configuration explanation:

- `interface gigabitEthernet 0/0` selects the physical router interface.
- `no shutdown` enables the physical interface.
- `interface gigabitEthernet 0/0.10` creates the subinterface for VLAN 10.
- `encapsulation dot1Q 10` associates the subinterface with VLAN 10.
- `ip address` assigns the gateway address.
- The same process is repeated for VLAN 20.

IEEE 802.1Q tagging allows the router to identify which VLAN each frame belongs to when traffic crosses the trunk link.

7.3 End-Device Configuration

Each PC was assigned a static IPv4 address through:

Desktop → IP Configuration

The IP addresses, subnet masks and default gateways were configured according to the addressing table in Section 5.

8. Network Testing and Verification

Connectivity was tested using the Command Prompt available on the Packet Tracer PCs.

The ICMP ping utility was used to verify communication between devices.

Test cases

Test	Source	Destination	Expected result
1	HR-PC1	192.168.10.11	Successful
2	IT-PC1	192.168.20.11	Successful
3	HR-PC1	192.168.20.10	Successful
4	IT-PC1	192.168.10.10	Successful

Reported outcome: All connectivity tests were successful.

This confirmed communication between devices within the same VLAN and across the two VLANs.

Verification commands

The following commands can be used to inspect the network configuration:

```
None
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
```

These commands help verify VLAN membership, trunk operation, interface status and the router's routing table.

9. Troubleshooting and Observations

During switch configuration, an invalid command was initially entered while assigning VLAN 20 to the relevant access ports.

The Cisco IOS CLI returned an invalid input indicator. The command was corrected, and the VLAN assignment was subsequently entered successfully.

The incident demonstrated the importance of checking command syntax and understanding the correct IOS configuration context.

The switch configuration was saved successfully, with the CLI returning `[OK]` after the save operation.

10. Security Considerations

Although this project focused primarily on network segmentation and connectivity, the design provides a foundation for implementing additional network security controls.

Recommended improvements include:

- Restricting unnecessary communication between departmental VLANs using Access Control Lists (ACLs).
- Disabling unused switch ports.
- Implementing port security on access ports.
- Using secure administrative access methods.
- Configuring appropriate management VLAN controls.
- Maintaining regular configuration backups.

These improvements were identified as future enhancements and are not claimed as implemented in this project.

11. Challenges and Lessons Learned

The project provided practical exposure to several important networking concepts.

Key lessons include:

1. VLANs provide logical separation of devices within a switched network.
2. Access ports connect end devices to their assigned VLANs.
3. Trunk ports carry traffic for multiple VLANs over a single physical connection.
4. IEEE 802.1Q provides VLAN tagging.
5. Router-on-a-Stick allows multiple VLANs to communicate through one physical router interface.
6. Correct IP addressing and default gateway configuration are essential for successful communication.
7. CLI verification and connectivity testing are important parts of network implementation.

12. Conclusion

The Small Office Network Design project was successfully implemented and tested in Cisco Packet Tracer.

Two departmental VLANs were configured, switch access ports were assigned, and a trunk connection was established between the switch and router.

Router-on-a-Stick was implemented to facilitate communication between the HR and IT networks.

Successful connectivity tests confirmed that the simulated network achieved its primary objectives.

This project represents the first stage of an ongoing practical Network Engineering portfolio focused on developing configuration, troubleshooting, documentation and network design skills.

13. Future Improvements

Potential extensions to this project include:

- Implementing DHCP for automatic IP address assignment.
- Applying ACLs to control inter-departmental communication.
- Adding a dedicated management VLAN.
- Introducing a server subnet.
- Configuring network monitoring using SNMP and Syslog.
- Expanding the design to support additional departments and network devices.

Project status: Completed.